

Assignment 3: IEEE 754 Floating-Point Representation and Arithmetic

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Deadline: two weeks from the date of issue — late submissions lose 10% per day unless a documented excuse is provided in advance (see the course policy in the syllabus).

Answer every question. Show all working for Numerical and Design questions — a bare final answer without the derivation earns partial credit only.

3.1 Conceptual

Q1. The true exponent of a floating-point number can be negative, and IEEE 754 stores it *biased* ($E_{\text{stored}} = E_{\text{true}} + 127$ for single precision) rather than in two's complement. Explain why the bias is used: what property of ordering the biased representation gives the hardware, and why that matters for comparison instructions. Your answer must state what a stored exponent of all zeros and all ones is *reserved* for.

Q2. A student writes a loop that accumulates $x = x + 0.1$ ten times and then tests $x = 1.0$. Explain why this test can fail even though $0.1 \times 10 = 1.0$ in exact arithmetic, and state the correct way to compare two floating-point numbers for “equality.”

3.2 Numerical

Q3. Encode the value 7.75 in IEEE 754 single precision (32 bits), showing: the binary expansion, the normalized scientific form $1.xxx_2 \times 2^E$, the sign bit, the biased exponent, and the 23-bit mantissa field. Write the final answer as an 8-hex-digit value (e.g. 0x412C0000).

Q4. Decode the 32-bit pattern 0x3F400000 into its decimal value, showing each field: sign, exponent, mantissa. Check your work by confirming the value against its binary scientific form. (Recall the lecture's anchor: 0x3F800000 = 1.0.)

3.3 Design

Q5. Perform the single-precision addition $10.5 + 0.25$ qualitatively following the four-step add algorithm (align, add, normalize, round). For each step state precisely what happens to the mantissas and exponents. Then repeat for $1\,000\,000.0 + 0.25$, and explain, in terms of the alignment step, why the 0.25 may be lost entirely — the same effect the lecture demonstrated with 0.1.

Q6. Derive the single-precision product of 1.5×2.0 using the multiplication rule (multiply mantissas, add exponents, subtract the bias once, normalize, round). State the exact product's

sign, exponent, and mantissa fields, and show that the result re-encodes to exactly 3.0. Explain why multiplication needs no alignment step.